

Xiangling Xu (许湘灵)

My Website

✉ xu[dot]xiangling[at]inria[dot]fr

G Google Scholar

ID Orcid



Research Interests: Quantum information, quantum foundations, operator algebras, noncommutative polynomials optimization

Education

- 2023 – now **Ph.D., Inria Saclay Île-de-France & École doctorale IP Paris**
Supervisor: Marc-Olivier Renou
Project: *Quantum correlations in causal structures: characterization and application*
- 2020 – 2023 **M.Sc. Mathematics, ETH Zürich**
Master Thesis: *Quantum Nonlocality in Bilocal Networks: An Operator Algebraic Perspective*
Semester Project: *Quantum BCOV Theory on Calabi-Yau Manifold*
CGPA: 5.69/6.00
- 2016 – 2020 **B.Sc. Mathematics and Physics, University of Toronto**
Projects: *Gravity with Noncommutative Geometry, Large-Scale Cosmological Structures, Optical Simulation, Biomechanical Simulation*
CGPA: 3.95/4.00

Publications and Preprints

- 1 F. M. Kalarde, S. Boreiri, **X. Xu**, L. Tendick, S. Beigi, P. Perinotti, T. Guaita, and M.-O. Renou, *Fermions are fundamentally more nonlocal than bosons*, Jun. 11, 2026. arXiv: 2606.12363 [quant-ph]. [URL: https://arxiv.org/abs/2606.12363](https://arxiv.org/abs/2606.12363).
- 2 **X. Xu**, M. Schötz, J. Wang, V. Magron, I. Klep, O. Fawzi, and M.-O. Renou, *The bulk spectral gap is semi-decidable: A convergent family of certified upper bounds*, Jun. 3, 2026. arXiv: 2606.03836 [quant-ph]. [URL: https://arxiv.org/abs/2606.03836](https://arxiv.org/abs/2606.03836).
- 3 F. M. Kalarde, **X. Xu**, and M.-O. Renou, *Comment on "quantum theory based on real numbers cannot be experimentally falsified": On the compatibility of physical principles with information theory for fermions*, Apr. 2026. arXiv: 2604.07425 [quant-ph]. [URL: https://arxiv.org/abs/2604.07425](https://arxiv.org/abs/2604.07425).
- 4 M.-O. Renou, **X. Xu**, and L. T. Ligthart, "Two convergent npa-like hierarchies for the quantum bilocal scenario," *Journal of Mathematical Physics*, vol. 67, no. 1, p. 012 203, Jan. 2026, ISSN: 0022-2488. [DOI: 10.1063/5.0211008](https://pubs.aip.org/aip/jmp/article-pdf/doi/10.1063/5.0211008/20865530/012203_1_5.0211008.pdf). eprint: https://pubs.aip.org/aip/jmp/article-pdf/doi/10.1063/5.0211008/20865530/012203_1_5.0211008.pdf.
- 5 M. Baroni, I. Klep, D. Leichtle, M.-O. Renou, I. Šupić, L. Tendick, and **X. Xu**, *Quantitative quantum soundness for all multipartite compiled nonlocal games*, 2025. arXiv: 2509.25145 [quant-ph]. [URL: https://arxiv.org/abs/2509.25145](https://arxiv.org/abs/2509.25145).
- 6 I. Klep, C. Paddock, M.-O. Renou, S. Schmidt, L. Tendick, **X. Xu**, and Y. Zhao, *Quantitative quantum soundness for bipartite compiled bell games via the sequential npa hierarchy*, 2025. arXiv: 2507.17006 [quant-ph]. [URL: https://arxiv.org/abs/2507.17006](https://arxiv.org/abs/2507.17006).
- 7 **X. Xu**, M.-O. Renou, and I. Klep, *Quantitative tsirelson's theorems via approximate schur's lemma and probabilistic stampfli's theorems*, 2025. arXiv: 2505.22309 [quant-ph]. [URL: https://arxiv.org/abs/2505.22309](https://arxiv.org/abs/2505.22309).

- 8 X. Luo, **X. Xu**, and X. Wang, “On the kinematic morphology around haloes,” *Monthly Notices of the Royal Astronomical Society*, vol. 518, no. 4, pp. 6059–6064, 2023.  DOI: <https://doi.org/10.1093/mnras/stac3500>.
- 9 **X. Xu**, “Quantum nonlocality in bilocal networks: An operator algebraic perspective,” M.S. thesis, ETH Zurich, 2023.  DOI: <https://doi.org/10.3929/ethz-b-000613670>.

Talks & Conference Activities

Invited Talks

- Jun 2026  *2026 SIAM Conference on Optimization (OP26)*, Edinburgh, United Kingdom
- Dec 2025  *Quantum Meets*, online at <https://quantum-meets.github.io>
- Jul 2025  *IWOTA 2025 Twente*, Twente, Netherland
- Oct 2023  *Saclay Quantum Seminar*, Inria Saclay, France
- Jun 2025  *IQC-PCQT-Quantum Saclay Workshop*, Paris, France

Contributed Talks

- Aug 2026  *QCrypt 2026*, Ottawa, Canada
- Jan 2026  *QIP 2026 Riga* – 2 contributed talks, Riga, Latvia
- Nov 2025  *3rd Colloquium GDR TeQ*, Grenoble, France
- Oct 2025  *YQIS25*, Barcelona, Spain
- Sep 2024  *Causalworlds 2024*, Waterloo, Canada
- Nov 2023  *Workshop Defi EQUIP*, Lyon, France

Talks during group visits

- Sep 2026  *Curiosity*, Inria & Télécom Paris & IP Paris
- Jul 2026  *Penghui Yao*, Nanjing University
- Feb 2026  *Jean-Daniel Bancal*, CEA, Institut de physique théorique
- Dec 2025  *Jie Wang*, AMSS-CAS, Beijing
- Nov 2025  *Thomas Vidick*, EPFL
- Oct 2025  *Jens Eisert*, Freie Universität Berlin
- Feb 2025  *QAT Team*, ENS Paris
- Jul 2024  *William Slofstra*, University of Waterloo
- Dec 2023  *David Gross*, University of Cologne

Poster Sessions

- Nov 2024  *2nd Colloquium GDR TeQ*, Paris, France
-  *YQIS24*, Paris, France

Other Participation

- Mar 2025  *828. WE-Heraeus-Seminar: Operator Theory and Polynomial Optimization in Quantum Information Theory*, Bad Honnef, Germany
- Sep 2024  *VQF Vienna Quantum Foundations Conference*, Vienna, Austria
- Nov 2023  *POP23 Workshop on Future Trends in Polynomial OPTimization*, Toulouse, France

Collaborations

Invited Group Visits

Oct-Nov 2026	David Gross, University of Cologne Invited three-week visit
Jul 2026	Penghui Yao, Nanjing University Invited one-week visit
Dec 2025 & May 2026	Jie Wang, AMSS-CAS, Beijing Two invited one-week visits
Oct-Nov 2025	Thomas Vidick, EPFL Invited two-week visit
Sep 2024	Igor Klep, University of Ljubljana Invited two-week visit funded by Erasmus+ Mobility program
Jul 2024	William Slofstra, Institute for Quantum Computing, University of Waterloo Invited two-week visit
Dec 2023	David Gross, University of Cologne Invited one-week visit

Organizations

Mar 2025 - Now	Coorganizer for <i>Seminars in Commutative and Noncommutative Polynomial Optimization for Quantum Theory</i> Coordinating the Gaspard Monge Visiting Professor program for Igor Klep Co-planning the seminar and the schedule for invited speakers
Mar 2025	Coorganizer for 828. <i>WE-Heraeus-Seminar at Bad Honnef</i> Exploring Operator Theory and Polynomial Optimization in Quantum Information Theory with world-renowned experts Coordinating with the organizers and inviting some of the speakers
Feb 2024 - Now	Coorganizer of <i>Project COMPUTE for Quanterra</i> Coordinating with the members Making the project poster and website
Sep 2023	Coorganizer for <i>Saclay Quantum Seminar on NPA hierarchy</i> Planning and organizing the one-day program to bring all quantum groups at Saclay together

Awards and Scholarships

2026	Erasmus+ Mobility Program, for visiting the University of Cologne
2024	Erasmus+ Mobility Program, for visiting the University of Ljubljana
2020	McNab Undergraduate Scholarship, University of Toronto
2017 - 2020	Dean's List Scholar, University of Toronto
2018	The Class of 3To and Associates Scholarship in Mathematics and Physics, University of Toronto
	International Exchange Award, University of Toronto
	Bronze Medal, 2017 University Physics Competition
2017	Eileen Doman Award, University of Toronto
	Summer Undergraduate Research Program, Canadian Institute for Theoretical Astrophysics
2016	President's Scholar of Excellence Program, University of Toronto

Teaching Experience

Lecture

- Dec 2025  *Recent trends in parallel, distributed, and quantum computing*
QDCS Master, Université Paris-Saclay
- Nov 2024  *Recent trends in parallel, distributed, and quantum computing*
QDCS Master, Université Paris-Saclay

Teaching Assistance

- 2022-2023  *Linear Algebra I & II*
Department of Mathematics, ETH Zürich
- 2021-2022  *Analysis I & II*
Department of Mathematics, ETH Zürich

Skills

- Languages  Chinese (native), English (fluent), German (beginner)
- Coding  Python (Ncpolzdpa), Matlab (Yalmip), Mathematica, C Language, Hugo on GitHub, $\text{\LaTeX}$

Referees

Available on Request